

Laboratory work 4

1

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'public' schema and the 'airline' table. The main query editor contains the following SQL query:

```
1 SELECT UPPER(airline_name) AS airline_name
2 FROM airline;
3
```

The Data Output pane at the bottom shows the results of the query:

airline_name
1 AIR ASTANA
2 FLYARYSTAN
3 LUFTHANSA
4 TURKISH AIRLIN...

The status bar indicates 'Total rows: 11' and 'Query complete 00:00:00.146'.

2

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, including the 'public' schema and the 'airline' table. The main query editor contains the following SQL query:

```
1 --1
2 SELECT UPPER(airline_name) AS airline_name
3 FROM airline;
4
5 --2
6 SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
7 FROM airline;
8
9
```

The Data Output pane at the bottom shows the results of the query:

airline_name
1 AIR ASTANA
2 FLYARYSTAN
3 LUFTHANSA
4 TURKISH AIRLIN...

The status bar indicates 'Total rows: 11' and 'Query complete 00:00:00.155'.

3

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (16)
 - airline
 - airline_3nf
 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_1
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook_db
- postgres
- snake_game

Query Editor: airport_db/postgres@PostgreSQL 17

```

1  --1
2  SELECT UPPER(airline_name) AS airline_name
3  FROM airline;
4
5  --2
6  SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
7  FROM airline;
8
9  --3
10 SELECT flight_id, airline_id, departing_gate, arriving_gate
11 FROM flights
12 WHERE airline_id IN (1, 2);
13
14
15

```

Data Output

flight_id	airline_id	departing_gate	arriving_gate
[PK] integer	integer	text	character varying (50)
1	1	A1	B2
2	2	C1	D2

Showing rows: 1 to 2 | Page No: 1 of 1

Total rows: 2 | Query complete 00:00:00.240 | CRLF | Ln 9, Col 1

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File Object Tools Edit View Window Help

Object Explorer

- Foreign Tables
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- Sequences
- Tables (16)
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 - booking_flight
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 - flight_3nf
 - flights
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- snake_game

Query Editor: airport_db/postgres@PostgreSQL 17

```

1  --1
2  SELECT UPPER(airline_name) AS airline_name
3  FROM airline;
4
5  --2
6  SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
7  FROM airline;
8
9  --3
10 SELECT flight_id, airline_id, departing_gate, arriving_gate
11 FROM flights
12 WHERE airline_id IN (1, 2);
13
14 --4
15 SELECT *
16 FROM airport
17 WHERE airport_name LIKE '%Regional%'
18 AND airport_name LIKE '%Air%';
19
20
21

```

Data Output

airport_id	airport_name	country	state	city	created_at	updated_at
[PK] integer	character varying (50)	character varying (50)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone

Total rows: 0 | Query complete 00:00:00.253 | CRLF | Ln 15, Col 1

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pgAdmin 4

Object Explorer

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- Sequences
- Tables (16)
 - airline
 - airline_3nf
 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
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- Types
- Views
- Subscriptions
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- postgres
- snake_game

Query Window: airport_db/postgres@PostgreSQL 17

```

--2
SELECT REPLACE(airline_name, 'Air', 'Aero') AS airline_name
FROM airline;

--3
SELECT flight_id, airline_id, departing_gate, arriving_gate
FROM flights
WHERE airline_id IN (1, 2);

--4
SELECT *
FROM airport
WHERE airport_name LIKE '%Reginal%'
AND airport_name LIKE '%Air%';

--5
SELECT
    first_name || ' ' || last_name AS full_name,
    TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
FROM passengers;
  
```

Data Output

full_name	birth_date
Albek Nurgaliyev	March 15, 1995
Aruzhan Saparova	July 22, 1998
Dias Kassymov	November 10, 19...
Madina Omarova	January 30, 2000

Total rows: 5 Query complete 00:00:00.280

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pgAdmin 4

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 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook_db
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- snake_game

Query Window: airport_db/postgres@PostgreSQL 17

```

--4
SELECT *
FROM airport
WHERE airport_name LIKE '%Reginal%'
AND airport_name LIKE '%Air%';

--5
SELECT
    first_name || ' ' || last_name AS full_name,
    TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
FROM passengers;

--6
SELECT flight_id, sch_arrival_time, act_arrival_time
FROM flights
WHERE act_arrival_time > sch_arrival_time;
  
```

Data Output

flight_id	sch_arrival_time	act_arrival_time
1	2026-09-21 09:30:00	2026-09-21 09:35:00
2	2026-09-21 13:00:00	2026-09-21 13:10:00
3	2026-09-22 10:30:00	2026-09-22 10:35:00
4	2026-09-22 17:00:00	2026-09-22 17:10:00

Total rows: 5 Query complete 00:00:00.372

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File Object Tools Edit View Window Help

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 - airport
 - airport_3nf
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 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
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- phonebook_db
- postgres
- snake_game

Query History

```

18 AND airport_name LIKE '%Air%';
19
20 --5
21 SELECT
22     first_name || ' ' || last_name AS full_name,
23     TO_CHAR(date_of_birth, 'Month DD, YYYY') AS birth_date
24 FROM passengers;
25
26 --6
27 SELECT flight_id, sch_arrival_time, act_arrival_time
28 FROM flights
29 WHERE act_arrival_time > sch_arrival_time;
30
31 --7
32 SELECT
33     flight_id,
34     sch_arrival_time,
35     act_arrival_time
36 FROM flights
37 WHERE act_arrival_time > sch_arrival_time;
38

```

Data Output

flight_id [PK] integer	sch_arrival_time timestamp without time zone	act_arrival_time timestamp without time zone
1	2026-09-21 09:30:00	2026-09-21 09:35:00
2	2026-09-21 13:00:00	2026-09-21 13:10:00
3	2026-09-22 10:30:00	2026-09-22 10:35:00
4	2026-09-22 17:00:00	2026-09-22 17:10:00

Total rows: 5 Query complete 00:00:00.381 CRLF Ln 32, Col 1

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File Object Tools Edit View Window Help

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 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
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- snake_game

Query History

```

25
26 --6
27 SELECT flight_id, sch_arrival_time, act_arrival_time
28 FROM flights
29 WHERE act_arrival_time > sch_arrival_time;
30
31 --7
32 SELECT
33     flight_id,
34     sch_arrival_time,
35     act_arrival_time
36 FROM flights
37 WHERE act_arrival_time > sch_arrival_time;
38
39 --8
40 SELECT *
41 FROM airline
42 WHERE airline_country IN ('France', 'Portugal', 'Poland')
43 AND created_at >= '2023-11-01'
44 AND created_at <= '2024-03-31';
45

```

Data Output

airline_id [PK] integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
----------------------------	--	--	---	---	---

Total rows: 0 Query complete 00:00:00.374 CRLF Ln 40, Col 1

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pgAdmin 4

Object Explorer

- Foreign Tables
- Functions
- Materialized Views
- Operators
- Procedures
- Sequences
- Tables (16)
 - airline
 - airline_3nf
 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook_db
- postgres
- snake_game

Query History

```

36 FROM flights
37 WHERE act_arrival_time > sch_arrival_time;
38
39 --8
40 SELECT *
41 FROM airline
42 WHERE airline_country IN ('France', 'Portugal', 'Poland')
43 AND created_at >= '2023-11-01'
44 AND created_at <= '2024-03-31';
45
46 --9
47 SELECT *
48 FROM baggage
49 WHERE weight_in_kg > 25
50 ORDER BY weight_in_kg DESC
51 LIMIT 3;
52
53
54
55
56

```

Data Output

baggage_id	weight_in_kg	created_at	updated_at	booking_id
[PK] integer	numeric (4,2)	timestamp without time zone	timestamp without time zone	integer

Total rows: 0 Query complete 00:00:00.370

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Object Explorer

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- Tables (16)
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 - airline_3nf
 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
- Views
- Subscriptions
- phonebook_db
- postgres
- snake_game

Query History

```

41 FROM airline
42 WHERE airline_country IN ('France', 'Portugal', 'Poland')
43 AND created_at >= '2023-11-01'
44 AND created_at <= '2024-03-31';
45
46 --9
47 SELECT *
48 FROM baggage
49 WHERE weight_in_kg > 25
50 ORDER BY weight_in_kg DESC
51 LIMIT 3;
52
53 --10
54 SELECT
55     first_name || ' ' || last_name AS full_name
56 FROM passengers
57 WHERE date_of_birth = (
58     SELECT MAX(date_of_birth)
59     FROM passengers
60 );
61

```

Data Output

full_name
Madina Omar...

Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.241

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pgAdmin 4

Object Explorer

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 - airport
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 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
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- Subscriptions
- phonebook_db
- postgres
- snake_game

Query History

```

49 WHERE weight_in_kg > 25
50 ORDER BY weight_in_kg DESC
51 LIMIT 3;
52
53 --10
54 SELECT
55     first_name || ' ' || last_name AS full_name
56 FROM passengers
57 WHERE date_of_birth = (
58     SELECT MAX(date_of_birth)
59     FROM passengers
60 );
61
62 --11
63 SELECT
64     booking_platform,
65     MIN(ticket_price) AS cheapest_price
66 FROM booking
67 GROUP BY booking_platform;
68
69

```

Data Output

	booking_platform	cheapest_price
1	Travel Agency	517.50
2	Website	172.50
3	Mobile App	253.00

Total rows: 3 Query complete 00:00:00.376

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pgAdmin 4

Object Explorer

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 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - passenger_3nf
 - passengers
 - security_check
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Query History

```

53
54 --10
55 SELECT
56     first_name || ' ' || last_name AS full_name
57 FROM passengers
58 WHERE date_of_birth = (
59     SELECT MAX(date_of_birth)
60     FROM passengers
61 );
62
63 --11
64 SELECT
65     booking_platform,
66     MIN(ticket_price) AS cheapest_price
67 FROM booking
68 GROUP BY booking_platform;
69
70 --12
71 SELECT *
72 FROM airline
73 WHERE airline_code ~ '[0-9]';
74

```

Data Output

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone

Total rows: 0 Query complete 00:00:00.386

13

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'airline', 'airline_3nf', 'airport', 'airport_3nf', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_3nf', 'booking_flight', 'booking_passenger', 'flight_3nf', 'flights', 'passenger_3nf', 'passengers', 'security_check', 'Trigger Functions', 'Types', 'Views', 'Subscriptions', 'phonebook_db', 'postgres', and 'snake_game'. The main query window is titled 'airport_db/postgres@PostgreSQL 17' and contains the following SQL code:

```
--11
SELECT
    booking_platform,
    MIN(ticket_price) AS cheapest_price
FROM booking
GROUP BY booking_platform;

--12
SELECT *
FROM airline
WHERE airline_code ~ '[0-9]';

--13
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;
```

The 'Data Output' tab shows the results of the last query (SELECT * FROM airline ORDER BY created_at DESC LIMIT 5). The table has 6 columns: airline_id (PK) integer, airline_code character varying (30), airline_name character varying (50), airline_country character varying (50), created_at timestamp without time zone, and updated_at timestamp without time zone. The results are as follows:

airline_id (PK) integer	airline_code character varying (30)	airline_name character varying (50)	airline_country character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone
1	10	KAZ	KAZAIR	2026-09-30 12:25:45.169758	2026-09-30 12:25:45.169758
2	9	FF	FLYFLY	2026-09-27 22:49:03.913696	2026-09-27 22:49:03.913696
3	7	AE	AIREASY	2026-09-27 22:49:03.913696	2026-09-27 22:49:03.913696
4	8	FH	FLYHIGH	2026-09-27 22:49:03.913696	2026-09-27 22:49:03.913696

14

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including tables like 'airline', 'airline_3nf', 'airport', 'airport_3nf', 'baggage', 'baggage_check', 'boarding_pass', 'booking', 'booking_3nf', 'booking_flight', 'booking_passenger', 'flight_3nf', 'flights', 'passenger_3nf', 'passengers', 'security_check', 'Trigger Functions', 'Types', 'Views', 'Subscriptions', 'phonebook_db', 'postgres', and 'snake_game'. The main query window is titled 'airport_db/postgres@PostgreSQL 17' and contains the following SQL code:

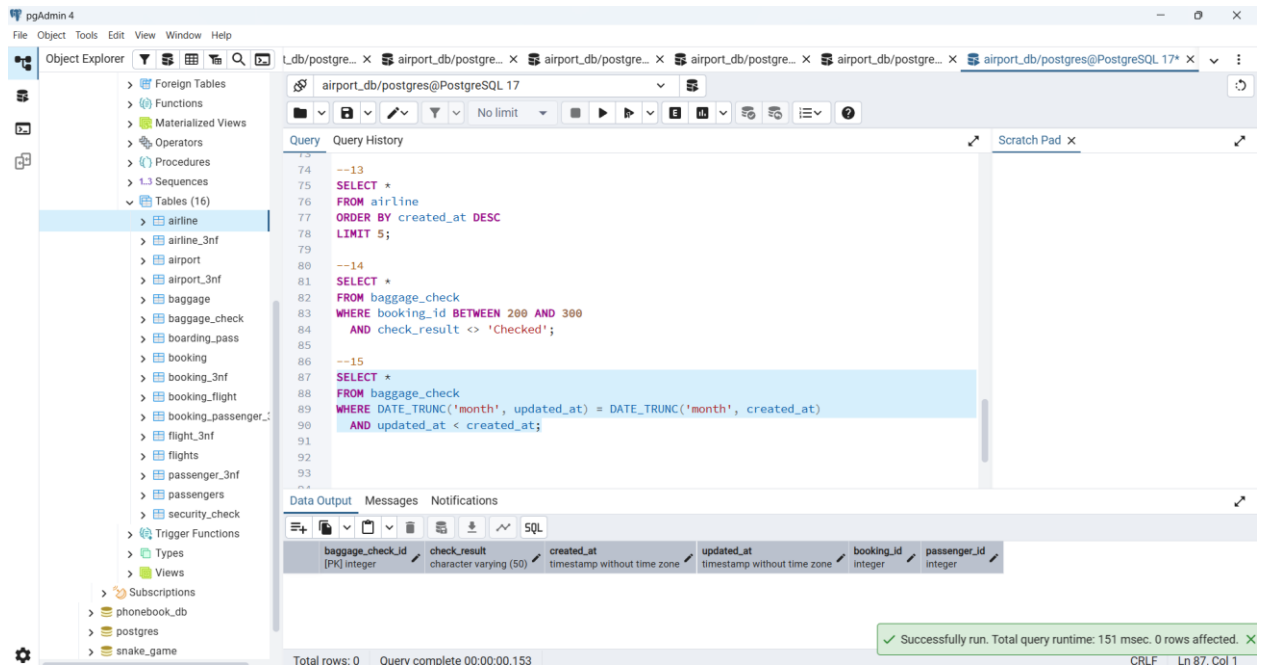
```
--13
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;

--14
SELECT *
FROM baggage_check
WHERE booking_id BETWEEN 200 AND 300
AND check_result <> 'Checked';
```

The 'Data Output' tab shows the results of the last query (SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300 AND check_result <> 'Checked'). The table has 6 columns: baggage_check_id (PK) integer, check_result character varying (50), created_at timestamp without time zone, updated_at timestamp without time zone, booking_id integer, and passenger_id integer. The results are as follows:

baggage_check_id (PK) integer	check_result character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	booking_id integer	passenger_id integer
-------------------------------	-------------------------------------	--	--	--------------------	----------------------

Total rows: 0 Query complete 00:00:00.437



pgAdmin 4

File Object Tools Edit View Window Help

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 - airport
 - airport_3nf
 - baggage
 - baggage_check
 - boarding_pass
 - booking
 - booking_3nf
 - booking_flight
 - booking_passenger_3nf
 - flight_3nf
 - flights
 - passenger_3nf
 - passengers
 - security_check
- Trigger Functions
- Types
- Views

Subscriptions

- phonebook_db
- postgres
- snake_game

Query Editor

Query History

```
--13
SELECT *
FROM airline
ORDER BY created_at DESC
LIMIT 5;

--14
SELECT *
FROM baggage_check
WHERE booking_id BETWEEN 200 AND 300
AND check_result <> 'Checked';

--15
SELECT *
FROM baggage_check
WHERE DATE_TRUNC('month', updated_at) = DATE_TRUNC('month', created_at)
AND updated_at < created_at;
```

Data Output

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
------------------	--------------	------------	------------	------------	--------------

baggage_check_id [PK] integer

check_result character varying (50)

created_at timestamp without time zone

updated_at timestamp without time zone

booking_id integer

passenger_id integer

Total rows: 0 Query complete 00:00:00.153

Successfully run. Total query runtime: 151 msec. 0 rows affected.

CRLF Ln 87, Col 1